MCFT: Multimodal Contrastive Fusion Transformer for Classification of Hyperspectral Image and LiDAR Data

Yining Feng, Jiarui Jin, Yin Yin, Chuanming Song, and Xianghai Wang

***Abstract*— Multisource remote sensing (RS) image fusion leverages data from various sensors to enhance the accuracy and comprehensiveness of Earth observation. Notably, the fusion of hyperspectral (HS) images and light detection and ranging (LiDAR) data has garnered significant attention due to their complementary features. However, current methods predominantly rely on simplistic techniques such as weight sharing, feature superposition, or feature products, which often fall short of achieving true feature fusion. These methods primarily focus on feature accumulation rather than integrative fusion. The transformer framework, with its self-attention mechanisms, offers potential for effective multimodal data fusion. However, simple linear transformations used in feature extraction may not adequately capture all relevant information. To address these challenges, we propose a novel multimodal contrastive fusion transformer (MCFT). Our approach employs convolutional neural networks (CNNs) for feature extraction from different modalities and leverages transformer networks for advanced fusion. We have modified the basic transformer architecture and propose a double position embedding mode to make it more suitable for RS image processing tasks. We introduce two novel modules: feature alignment module and feature matching module, designed to exploit both paired and unpaired samples. These modules facilitate more effective cross-modal learning by emphasizing the commonalities within the same features and the differences between features from distinct modalities. Experimental evaluations on several publicly available HS–LiDAR datasets demonstrate that proposed method consistently outperforms existing advanced methods. The source code for our approach is available at: https://github.com/SYFYN0317/MCFT.**

***Index Terms*— Contrastive learning, deep learning (DL), feature alignment, feature matching, HS–LiDAR fusion and classification, vision transformer (ViT).**

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I. Introduction

With the proliferation of Earth observation platforms and remote sensing (RS) sensors, acquiring diverse RS data with distinct characteristics from the same geographical area has become increasingly feasible [1]. Hyperspectral (HS) image is notable for its continuous and extensive spectral information, broad surface coverage, and exceptional object detection capabilities [2]. In contrast, light detection and ranging (LiDAR) data provides critical elevation information, allowing for the differentiation of ground features with similar spectral signatures but varying heights [3]. The fusion of HS image and LiDAR data results in a more comprehensive dataset, enhancing the ability to identify unique ground object attributes and improving classification accuracy [4]. This fusion of HS and LiDAR data has become a key focus in land use/land cover (LULC) studies, with wide-ranging applications across various fields [5].

In the initial stages of research, multisource RS data fusion typically involves feature extraction, feature transformation, and weight-based fusion strategies. Ghamisi et al. [6] employed morphological operations on the first principal component analysis (PCA) of HS images and the digital surface model (DSM) of LiDAR data to derive spatial features from both datasets. Rasti et al. [7] optimized feature fusion by incorporating constraints, utilizing sparse low-rank methods for dimensionality reduction, and subsequently fusing the extracted features before classification. Liao et al. [8] used a graph-based model to extract and fuse features from HS images and LiDAR data. Rasti and Ghamisi [9] proposed an optimization problem targeting low-rank minimization, leveraging an iterative algorithm based on the multiplier alternating direction method to estimate fused features within a low-dimensional subspace. While traditional methods largely depend on feature extraction techniques, which help reduce data dimensions, their generalization capacity is limited, indicating a need for further research to enhance feature extraction accuracy from multisource data.

Recently, the field of RS image fusion and classification has seen growing interest in deep learning (DL) due to its advanced feature extraction capabilities [10]. Techniques integrating HS image and LiDAR data through DL have gained prominence, leveraging convolutional neural networks (CNNs) to autonomously extract complex semantic and hierarchical features. Salman and Yüksel [11] fused HS image with LiDAR

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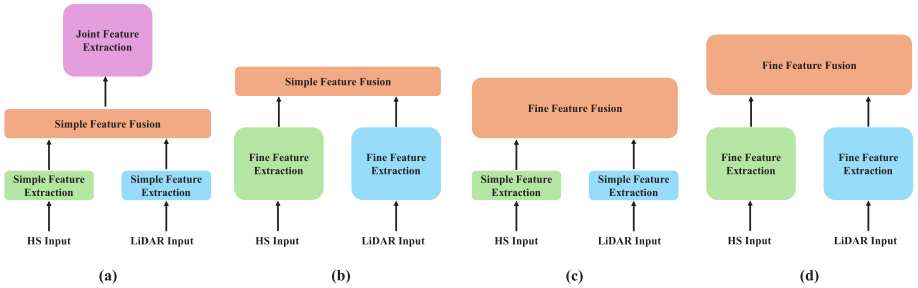


Fig. 1. Four categories of HS image and LiDAR data fusion models. (a) Simple feature extraction, simple fusion strategy, and joint feature extraction. (b) Fine feature extraction and simple fusion strategy. (c) Simple feature extraction and fine fusion strategy. (d) Fine feature extraction and fine fusion strategy.

data and employed the initial convolutional layer filter of AlexNet to extract elevation details from LiDAR data. However, single-branch input methods often struggle to preserve the unique information inherent in each multisource dataset. Xu et al. [12] introduced a two-branch CNN (TBCNN) that integrates spectral and spatial branches for joint classification of HS image and LiDAR data. Zhao et al. [13] proposed the fractional Gabor convolutional network (FGCN), focusing on efficient feature integration and comprehensive feature extraction. Zhang et al. [14] introduced the structurally optimized transmission network (SOT-Net) to enhance semantic correlation across multiple sources. Although multibranch input network structures can mitigate information loss from different modalities during the fusion process, they still fail to establish a connection between the spatial information of the two modalities. Wu et al. [15] developed the cross-channel reconstruction CNN (CCR-Net), which enhances the learning of fusion representations through cross-modal reconstruction, facilitating effective information exchange between modalities. Wang et al. [16] proposed a multiattention hierarchical dense fusion network (MAHiDFNet) that improves feature fusion efficacy by integrating a three-branch framework emphasizing spatial, spectral, and elevation information with an attention mechanism. Gao et al. [17] proposed an adaptive multiscale spatial–spectral enhancement network (AMSSE-Net). This approach not only facilitates a deeper exploration of the spectral–spatial information within the HS data but also effectively fuses this information with the elevation data obtained from LiDAR. Zhang et al. [18] proposed a novel cross-modal feature aggregation and enhancement network for the joint classification of HSI and LiDAR data. Beyond CNNs, other network architectures have also been employed for fusion and classification, including stacked autoencoders (SAEs) [19], generative adversarial networks (GANs) [20], and graph convolutional networks (GCNs) [21].

While DL techniques for the fusion and classification of HS image and LiDAR data have shown promising results, most existing methods predominantly utilize single-level fusion networks. These networks combine features from diverse sources at various network stages using straightforward mechanisms such as weight sharing, feature superposition, or product methods. However, this simplistic approach to fusion primarily focuses on feature accumulation rather than true feature integration. Consequently, the majority of fusion and classification methods for HS images and LiDAR data concentrate on feature extraction. As illustrated in Fig. 1, prevalent methods generally adopt configurations similar to (a) and (b), with an emphasis on feature extraction. Specifically, Fig. 1(a) depicts a method where joint fused features are extracted, whereas Fig. 1(b) shows a method where multisource features are extracted separately.

The transformer framework, emerging as a novel model, utilizes self-attention mechanisms to manage data effectively, thus facilitating the fusion of multimodal data. Recent studies have begun applying this framework within the RS domain. However, current approaches employing the transformer framework largely follow a configuration similar to Fig. 1(c), which involves extracting features and feeding them into a transformer model for fusion. This simplistic feature extraction approach may compromise the quality of extracted features and consequently, impact the effectiveness of fusion. Therefore, an optimal fusion framework should integrate both thorough feature extraction and more advanced fusion methodologies [as depicted in Fig. 1(d)] to maximize the utility of multimodal features. Although existing multimodal transformer enable joint modeling of multimodal tokens, alignment issues occur when features from different modalities are inputted into the transformer [22]. These features, residing in their distinct spaces, challenge multimodal encoders in learning and modeling their interactions accurately. Thus, learning the correlation relationships between modalities remains a significant challenge. Additionally, prevailing methodologies often focus on training with pairs of congruent labels, overlooking multimodal samples with disparate labels and neglecting potentially valuable "contrary information." Meanwhile, the universal transformer framework is proposed based on natural images, and the position embedding used is not suitable for RS image processing tasks that mainly use patches as inputs. Therefore, directly applying the transformer framework to multisource RS fusion classification tasks often fails to achieve satisfactory results.

To address these challenges, we propose a novel multimodal contrastive fusion transformer (MCFT) framework. This framework leverages CNNs for comprehensive feature extraction across various modalities and employs a transformer model, optimized for RS tasks, as an effective fusion mechanism to improve classification accuracy. Traditional approaches often face difficulties in aligning features from different modalities when input into the transformer, leading to challenges in learning and modeling their

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interactions accurately. The basic transformer framework has been improved by adopting a double position embedding module, which highlights the importance of the central pixel in the patch. Our proposed solution aligns multimodal features to overcome these difficulties. Additionally, recognizing the limitation of current methods that focus predominantly on paired samples during training—thereby underutilizing multimodal samples with dissimilar labels—we introduce a challenging negative sample contrastive learning approach. This approach aims to enhance network robustness and improve classification performance. The primary innovations of this study include the following:

1) A novel MCFT is proposed, where features from diverse modalities are extracted using CNNs and combined through a transformer network. Improvements have been made to the basic modules in the transformer architecture, proposing a double position embedding mode and an improved cross-attention module to enhance feature extraction and fusion processes, thereby improving classification accuracy.
2) A strategy for aligning multimodal features has been proposed, which addresses the issue of misalignment among different modal features when fed into multimodal transformer. A multimodal contrastive loss is introduced for embedding two modalities. This facilitates the alignment of representations for HS images and LiDAR data before fusion, thereby simplifying subsequent cross-modal learning by multimodal transformer.
3) A novel feature matching module has been proposed. Perform binary classification tasks by constructing the most challenging negative sample pairs and calculating the multimodal matching loss together with the positive sample pairs. By utilizing the information between different labeled samples, the problem of limited labeled sample size in multisource RS data processing has been effectively improved.
4) The proposed methodology is assessed using four publicly available real-world datasets, with performance compared against several state-of-the-art (SOTA) algorithms. Exceptional performance is demonstrated by the experimental results.

The subsequent structure of this article is as follows: Section II consolidates prior studies on vision transformer (ViT) and contrastive learning. In Section III, a comprehensive overview of the framework and novel modules proposed in this manuscript is presented. Section IV delineates the experimental configuration, results analysis, and ablation experiments. Lastly, Section V comprises the concluding remarks of this article.

II. Related Work

A. ViT for LULC Classification

Transformer, initially introduced by researchers in the domain of natural language processing (NLP) [23], has evolved significantly over time and now holds a pivotal position within the realm of NLP. It leverages the self-attention mechanism to effectively capture extensive feature dependencies within input sequences. This mechanism serves as the cornerstone of transformer, facilitating the analysis of interrelations among elements across varying distances in the input sequence. Essentially, the primary function of the self-attention mechanism entails discerning salient information from the input sequence, assigning it higher emphasis, while attenuating less significant details through appropriate weight adjustments. The self-attention mechanism can be expressed as follows:

$$\text{Attention}(Q,\ K,\ V)\ = \text{softmax}\left(\frac{QK^T}{\sqrt{d_q}}\right)V \tag{1}$$

where Q is the query, K is the key, V is the value, and d_q is the dimension of Q. Within the transformer framework, a frequently employed technique is the multihead attention mechanism, which serves to analyze input sequences by partitioning the model into distinct heads, thereby establishing multiple subspaces that enable focused exploration of diverse facets of information. This approach entails iteratively applying the self-attention process across multiple iterations and subsequently amalgamating the resulting outputs. The computational procedure is delineated in (2), where head_h represents the attention outcome from individual subspaces, and W^O represents the weight of heads

$$\text{MultHead}(Q,\ K,\ V)\ = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O. \tag{2}$$

The success of the transformer model in NLP has spurred researchers deeply involved in computer vision (CV) tasks to explore new avenues. Dosovitskiy et al. [24] proposed the ViT, marking the first application of the transformer architecture in CV. ViT demonstrated remarkable performance, indicating the potential of transformer-based frameworks to replace convolutional operations in deep neural networks for CV tasks.

The proliferation of transformer in CV has catalyzed researchers to explore their application across diverse tasks in RS image processing. In recent years, numerous researchers have employed the transformer framework for the fusion classification task. Yao et al. [25] proposed an innovative multimodal DL framework tailored for LULC classification tasks by extended ViT (ExViT). Ding et al. [26] devised the global–local transformer network (Glt-Net), which optimally combines the strengths of CNNs for local features and transformers for modeling long-range dependencies. Xue et al. [27] devised the deep hierarchical ViT (DHViT) architecture tailored for the unified classification of HS image and LiDAR data. Roy et al. [28] proposed a novel multimodal fusion transformer (MFT) network, integrating additional multimodal data as external classification (CLS) labels within the transformer encoder, facilitating enhanced generalization capabilities. Zhang et al. [29] presented a new model called multimodal transformer network (MTNet) that leverages transformer advantages to capture both the specific and shared characteristics of HS image and LiDAR data.

While transformer-based approaches have demonstrated favorable outcomes, they commonly rely on uncomplicated linear transformations for extracting multimodal features. The existing position embedding module is tailored for natural

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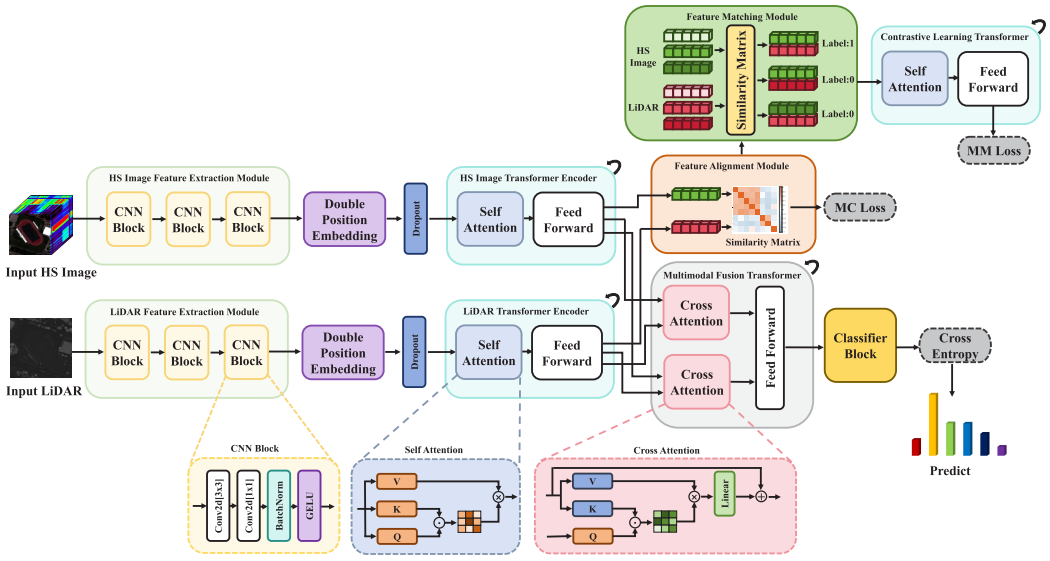


Fig. 2. Proposed MCFT framework for HS image and LiDAR fusion and classification.

images and has not been optimized for RS images. Moreover, the conventional cross-attention module fails to adequately incorporate information from both modalities, and the integrated modeling of multimodal labeling does not guarantee the comprehensive exploitation of multimodal features. Simultaneously, disparate modal features inputted into the transformer remain unaligned, with embedded features residing in separate spaces. Consequently, this complicates the learning and modeling of interactions by the multimodal encoder, leading to challenges for the multimodal transformer in accurately capturing inter-modality correlations. These issues may impede the full exploitation of the transformer framework's advantages.

B. Contrastive Learning

Contrastive learning is a machine learning method that learns by comparing features or attributes between two or more different entities [30], [31]. This method is commonly used to solve problems such as classification, regression, and clustering. In contrastive learning, the system attempts to learn knowledge from the similarities and differences between entities, rather than directly learning from labeled training data. This makes contrastive learning particularly useful in situations where labeled data is scarce or difficult to obtain. Using contrastive learning for feature alignment is a method aimed at aligning or mapping feature spaces from different data distributions to a shared feature space [32].

At present, researchers have applied contrastive learning to the field of RS and achieved certain results. Hou et al. [33] proposed an HS image classification algorithm grounded in contrastive learning principles. The approach leverages the data from ample unlabeled samples to mitigate the challenge of inadequate label information in HS datasets. Li et al. [34] proposed a global style and local matching contrastive learning network (GLCNet) for RS image semantic segmentation. Liu et al. [35] proposed a refined prototypical contrastive learning network for few-shot learning (RPCL-FSL) for small-sample HS image classification. Li et al. [36] proposed a classification method based on pseudo-labeled contrastive learning to exploit substantial amounts of unlabeled information in order to enhance intra-class information interaction. Ding et al. [37] proposed a novel uncertainty-aware contrastive learning (UACL) method. By jointly learning from both labeled and unlabeled samples, more discriminative spectral–spatial feature representation will lead to a further boost in classification performance.

In summary, the contrastive learning has been widely applied, but there are still some shortcomings. In multimodal RS data, there may be an imbalance in the quantity of different types of data, which can lead to the contrastive learning models leaning toward larger data types in the learning process and ignoring smaller data types. Meanwhile, the contrastive learning models typically learn feature representation consistency within the same data type, but, in multimodal problems, the consistency between different data types needs to be considered, which increases the complexity of the model design.

III. Methodology

A. Overall Architecture

The overall proposed framework is shown in Fig. 2. The network consists of four parts as a whole.

1) Feature extraction module based on CNNs. Use CNNs to extract fine features from the input image.
2) Feature fusion module based on transformer. Improve the traditional position embedding and cross-attention mechanism to fully fuse and represent multimodal features.
3) Multimodal feature alignment module. By calculating the similarity of each multimodal feature, the features are aligned, which is beneficial for subsequent multimodal fusion.
4) Multimodal feature matching module based on contrastive learning. By constructing the most challenging

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negative sample strategy and fully utilizing unmatched training samples, the goal of comprehensive information utilization is achieved.

B. Feature Extraction Module Based on CNNs

The ViT model was originally developed for natural image classification in CV, and it operates in a task environment that is significantly different from RS-based LULC classification. Pre-training on a wide range of datasets such as ImageNet in natural image environments can significantly improve the performance of transformer models, enabling them to effectively capture salient patterns at both global and regional levels. However, in RS tasks, the lack of large-scale datasets for pre-training exacerbates ViT's inherent requirement for a large number of pre-training samples. In addition, the recognition of image level semantics mainly depends on capturing salient patterns of the global or regional context, rather than delving into complex inter-pixel information. On the contrary, LULC classification mainly focuses on the classification attributes of individual pixels within a given scene. Given the characteristics of land features and the limited availability of training data in RS applications, it is crucial to utilize spatial contextual information in our network design. CNNs are naturally adept at capturing local relationships, which makes them particularly advantageous when training samples are scarce. By integrating CNN with transformer architecture, we can effectively combine the strengths of these two frameworks to address their respective weaknesses. This hybrid method is expected to optimize the performance of LULC classification tasks and achieve excellent results.

The CNN blocks play a crucial role in the initial feature extraction for HS image and LiDAR data by leveraging their ability to capture spatial and spectral information effectively. They apply convolution operations to the input data, allowing the network to learn spatial hierarchies and local patterns. For HS data, this means extracting spectral features from multiple bands; while, for LiDAR data, it involves capturing spatial and elevation relationships. As the same time, CNNs can process multichannel inputs effectively. In HS image, each pixel has multiple spectral bands, while LiDAR data also includes various attributes. CNN blocks can handle these multidimensional inputs seamlessly. The output from CNN blocks consists of feature maps that highlight important features relevant for fusion and classification tasks. These maps serve as the foundation for subsequent processing stages. Formally, given the input HS image and LiDAR patches, denoted as $\{x_i\}_{i=1}^{N} \in \mathbb{R}^C$, $\{d_i\}_{i=1}^{N} \in \mathbb{R}^C$, extract multimodal features using the following formula:

$$h_i^{(e)} = \begin{cases} f\left(\mathrm{BN}_{\gamma,\beta}\left(\omega_h^{(e)T} x_i + b_h^{(e)}\right)\right), & e = 1 \\ f\left(\mathrm{BN}_{\gamma,\beta}\left(\omega_h^{(e)T} h_i^{(e-1)} + b_h^{(e)}\right)\right), & e = 2, 3 \end{cases} \tag{3}$$

$$l_i^{(e)} = \begin{cases} f\left(\mathrm{BN}_{\gamma,\beta}\left(\omega_l^{(e)T} d_i + b_l^{(e)}\right)\right), & e = 1 \\ f\left(\mathrm{BN}_{\gamma,\beta}\left(\omega_l^{(e)T} l_i^{(e-1)} + b_l^{(e)}\right)\right), & e = 2, 3 \end{cases} \tag{4}$$

where h_i and l_i represent the eth features extracted by the encoder block of HS image and LiDAR data. $\{\omega_h^{(e)}, b_h^{(e)}\}$ and $\{\omega_l^{(e)}, b_l^{(e)}\}$ are sets of weights and biases for the encoder parts of multimodal data. $f(\cdot)$ is the GELU nonlinear activation function [38]. Selecting GELU as the activation function facilitates gradient propagation during the training phase of deep neural networks, mitigating issues associated with the discontinuous derivatives of the ReLU function at 0 and alleviating the problem of vanishing gradients. Additionally, the GELU function introduces a transformation akin to the sigmoid function in the nonlinear activation function, enabling the GELU function's output to span a broader range, thereby expediting the model's convergence rate. $\mathrm{BN}_{\gamma,\beta}$ represents batch normalization (BN) layer. Its function is to accelerate parameter learning and avoid gradient vanishing during training process [39].

C. Feature Fusion Module Based on Transformers

Similar to ViT dividing images into non-overlapping patches, we map the output features of the convolution module to P^2 vectors, each representing a pixel within the patch. Then, these embedded features are processed according to tokenization rules, and a series of 1-D tokens $\mathbf{z}_i \in \mathbb{R}^{P^2 \times d}$ are obtained. Subsequently, linear layers are established for distinct modalities, aiming to map their extracted features onto a shared subspace. By amalgamating the aforementioned procedures into a comprehensive trainable function, denoted as $g(\cdot)$, the tokenized representation of the ith patch is acquired as delineated below:

$$\mathbf{z}_i = \mathrm{DP}\left(\left[g\left(x_{i,1}\right); g\left(x_{i,2}\right); \ldots; g\left(x_{i,P^2}\right)\right] + e\right) \tag{5}$$

where DP represents the dropout layer to prevent overfitting issues in our LULC classification network. $e \in \mathbb{R}^{(P^2+1)\times d}$ is a learning position embedding added to the token to preserve position information. In order to better align multimodal tokens and reduce the solution space, we consider embedding e into the same block with shared positions for different modalities.

In contrast to traditional image classification tasks in CV, LULC classification often utilizes individual pixels as the fundamental unit for categorizing land cover types. This approach aims to ascertain the land cover category associated with each pixel. However, focusing exclusively on single pixels can result in the loss of spatial continuity. To address this issue, the pixels requiring classification are typically designated as the central pixel within an appropriately sized input patch block. Thus, the role of the central pixel is critical in LULC classification tasks [40]. According to the first law of geography, there exists a significant correlation between the land cover classification of the central pixel and that of its neighboring pixels. Previously, ViT-based methods have predominantly employed positional embeddings analogous to those used in traditional image classification tasks. This method strives to represent the spatial position of individual pixels within a patch through absolute positional representation, as illustrated in Fig. 3(a). While this approach effectively captures spatial information, it fails to emphasize the unique significance of the central pixel, which can lead to misclassification of isolated pixels as belonging to the surrounding background.

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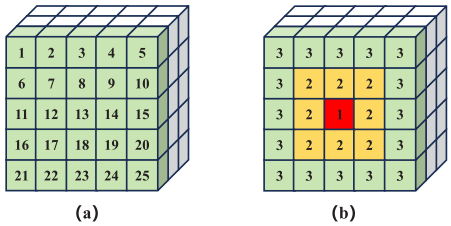


Fig. 3. Different position embedding modes. (a) Traditional position embedding. (b) Center position embedding.

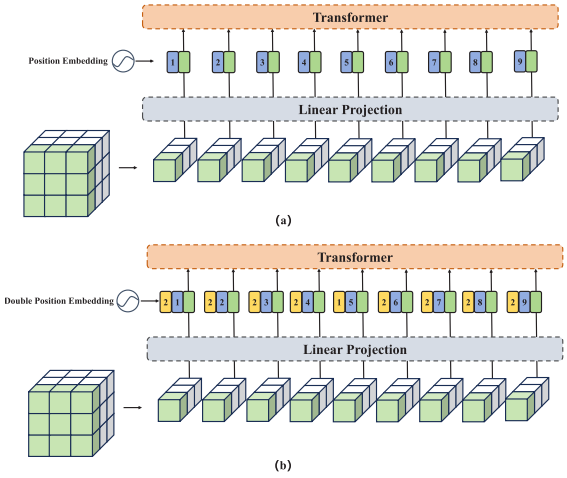


Fig. 4. Comparison of different ViTs. (a) ViT using traditional positional embedding. (b) ViT using double positional embedding.

To overcome this limitation, we have designed a novel center position embedding pattern, as depicted in Fig. 3(b). This center position embedding mode prioritizes the central pixel by treating it as a focal point and encoding the surrounding context sequentially outward. This adjustment enhances the model's capacity to accurately reflect the importance of the central pixel in relation to its neighboring pixels. The expression is as follows:

$$e + \text{Pixel}(i, j) = e_{\min(|i-i_c|,|j-j_c|)+1} = e_k \quad k = 0, 1, \ldots, (w/2+1) \tag{6}$$

where i and j represent pixel column and row indices within the input patch, while i_c and j_c denote the column and row indices associated with the central pixel, and w signifies the patch size.

To mitigate this limitation, as illustrated in Fig. 4, we have devised a double position embedding mode that incorporates both absolute spatial position and highlights the distinctive position of the central pixel.

Unlike traditional ViT, it is not always necessary to consider CLS tokens because each token is processed equally, and simply ignoring these tokens may lead to inevitable information loss. Therefore, in the proposed MCFT, we suggest fusing all tokens to obtain the final token for classification, rather than introducing additional CLS for each modality. After tokenizing the features, we use two transformer networks with different parameters to extract features from different modalities, which use a universal multihead self-attention module internally.

Cross-attention is a common mechanism in multimodal learning, which is usually based on self-attention mechanism, but it considers data from different sources when processing input. The information of states guides the processing of another modality [41]. For example, in tasks that combine images and text, cross-attention may be used to better guide the extraction of image features through text information or vice versa. This is an effective way to integrate different types of data. Recently, cross-attention mechanisms have garnered heightened interest in the research community [42]. In traditional cross-attention modules, queries come from a representation of one modality (such as text), and keys and values come from a representation of another modality (such as images). Calculate attention scores using query vectors and key vectors. Finally, the attention score is used to weigh and sum the value vectors. Through this approach, the model can better understand and integrate information from different modalities, thereby improving its ability to handle complex scenarios. However, it may result in the loss of some modality-specific information, especially in the field of RS, where each modality's data has its unique meaning.

To this end, we adopt a novel cross-attention method that generates Q, K, and V for the features of two modalities, and cross calculates the dot product of queries and keys. Firstly, project the token of HS modality T_H onto two independent matrices $V_H, K_H \in \mathbb{R}^{HW\times C}$, to calculate a set of values and keys. Then, project the token of LiDAR modality T_L onto another separate matrix $Q_L \in \mathbb{R}^{HW\times C}$ to compute a set of queries

$$V_H = T_H W^V, \quad K_H = T_H W^K, \quad Q_L = T_L W^Q \tag{7}$$

where W^V, W^K, and W^Q denote the weight matrices. Then, we apply the softmax function to obtain the attention score, and use the attention score to weighted sum the value vectors. This weighted sum represents the output that combines two types of modal information. The detailed information on the cross-attention module is as follows:

$$Z_H = \text{softmax}\left(\frac{Q_L {K_H}^T}{\sqrt{D_K}}\right) \cdot V_H \tag{8}$$

where we also adopt a multihead cross-attention mechanism comprising four parallel heads, facilitating the model's comprehensive comprehension of the interrelationship between the two modal features from diverse viewpoints. Thirdly, Z_H is reprojected back into the original space through nonlinear transformation and added to the input sequence through residual connections

$${T_H}' = \alpha \cdot Z_H W^O + \beta \cdot T_H \tag{9}$$

where W^O represents the output weight matrix before the feedforward network (FFN) layer. Two learnable parameters, denoted as α and β, are initialized to 1 during training in order to adaptively learn information from different branches, thereby enhancing model performance. Subsequently, a two-layer fully connected FFN is employed to further refine global information, enhance model robustness and accuracy, and produce improved features.

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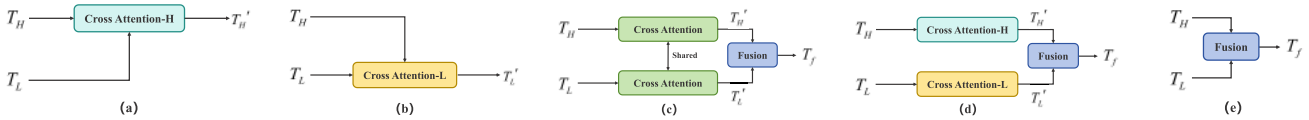


Fig. 5. Comparison of different fusion modes. (a) Single cross-attention mode for HS image. (b) Single cross-attention mode for LiDAR. (c) Dual cross-attention mode (shared) for both modalities. (d) Dual cross-attention mode for both modalities. (e) Without cross-attention mode.

In parallel, we investigate five fusion modes to evaluate the effectiveness of the cross-attention module, as illustrated in Fig. 5. Fig. 5(a) and (b) illustrates the traditional cross-attention mode, where only one modality feature is printed, with the feature of one modality integrated as a complementary aspect into another modality. Fig. 5(c) and (d) presents two distinct operational modes of the dual cross-attention module utilizing shared and non-shared parameters. Fig. 5(e) represents the baseline feature fusion method. Currently, most methods primarily employ cross-attention mechanisms, such as those depicted in structures represented in Fig. 5(a) or (c), to fuse multisource data. However, both approaches exhibit certain limitations. Structure represented in Fig. 5(a) heavily relies on the features of HS images as the primary source, which diminishes the effectiveness of LiDAR data as a supplementary source. Conversely, structure in Fig. 5(c) adopts a shared weight mechanism that treats the features of both modalities equally, which can lead to the spatial information from LiDAR adversely influencing the ground information discernible through spectral features. Therefore, this study proposes structure in Fig. 5(d) as a novel cross-attention framework. This structure facilitates a comprehensive understanding of the input data, promotes the integration of information from different modalities, enhances the representation of both modalities, and ultimately achieves fine-grained data fusion. Thanks to the cross-attention mechanism, our method can deliver satisfactory results even when one input modality exhibits poor image quality. Additionally, in the subsequent ablation experiments section, we conducted further investigations in this area and provided a more detailed analysis of our findings.

D. Multimodal Feature Alignment Module

Currently, the majority of methods for fusing and classifying HS images and LiDAR data involve direct fusion of features post feature extraction. However, as features from different modalities are situated in their respective subspaces, they encounter difficulty in interacting with one another. Therefore, ensuring feature alignment is crucial, as discrepancies in feature representations between different modality images can result in inconsistent or unreliable fusion features, a process often overlooked. Feature alignment enables the alignment of features from heterogeneous modalities to a shared feature space during the fusion process, thereby enhancing the consistency and reliability of the fused features. This approach enables different forms of information to better complement and collaborate within the fused features, ultimately improving the understanding and analytical capabilities of RS images. Additionally, feature alignment aids in reducing inter-modality differences, enhancing the model's ability to generalize to multimodal data, thereby strengthening the robustness and reliability of RS image processing tasks and facilitating subsequent cross-modal learning by multimodal transformers.

To this end, we calculate the similarity between the features of two modalities after passing through separate transformers, and then use a loss function to constrain the similarity, thereby achieving feature alignment before fusion. Firstly, perform linear transformations on the features of the two modalities F_H and F_L. Then perform matrix dot product to calculate the similarity between HS to LiDAR and LiDAR to HS:

$$\begin{cases} s(H, L) = g_v(F_H) \cdot g_w(F_L)^T \\ s(L, H) = g_W(F_L) \cdot g_v(F_H)^T \end{cases} \tag{10}$$

where $g_v(\cdot)$ and $g_w(\cdot)$ are linear transformations that map features to standardized low-dimensional representations. The dot product is a common method for calculating the similarity between two vectors. When two vectors perform dot product operations, the size of the result can reflect the similarity between the two vectors. This is because dot product operations consider the direction and size of vectors. Calculate the similarity as the soft similarity between two modal data. For each HS image and LiDAR, we calculate the softmax normalized similarity from HS image to LiDAR and from LiDAR to HS image as follows:

$$\begin{cases} p_m^{H2L}(H) = \dfrac{\exp(s(H, L)/\tau)}{\sum_{m=1}^{M} \exp(s(H, L)/\tau)} \\ p_m^{L2H}(L) = \dfrac{\exp(s(L, H)/\tau)}{\sum_{m=1}^{M} \exp(s(L, H)/\tau)} \end{cases} \tag{11}$$

where τ is the scaling factor to control the range and stability of similarity values, with the aim of avoiding the results being overly influenced by the features of a certain modality. Let $y^{H2L}(H)$ and $y^{L2H}(L)$ represent the ground-truth one-hot similarity, wherein negative pairs exhibit a probability of 0 and the positive pair exhibits a probability of 1. Calculate soft pseudo-labels through soft similarity and ground-truth similarity

$$\begin{cases} q^{H2L}(H) = \theta p^{H2L}(H) + (1-\theta) y^{H2L}(H) \\ q^{L2H}(L) = \theta p^{L2H}(L) + (1-\theta) y^{L2H}(L) \end{cases} \tag{12}$$

where θ is the balance parameter for adjusting the ratio of the two similarity. We consider HS images and LiDAR as different variants for capturing their semantics, with the goal of learning representations that are not affected by viewpoint changes. This can be achieved by maximizing the mutual information (MI) between the two features. In practice, we maximize MI(H, L) by minimizing the InfoNCE loss [43]. InfoNCE

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loss is a commonly used contrastive loss function in self-supervised learning tasks, especially in the framework of contrastive learning. It aims to narrow the distance between positive samples (similar samples) while pushing the distance between negative samples (dissimilar samples), in order to learn effective representations of data. Define the multimodal contrastive loss ($\mathscr{L}_{\mathrm{MC}}$) as

$$\mathscr{L}_{\mathrm{MC}} = -\frac{1}{2}\left(\sum_{m} q_m^{H2L}(H)\log p_m^{H2L}(H) + \sum_{m} q_m^{L2H}(L)\log p_m^{L2H}(L)\right). \tag{13}$$

Minimizing $\mathscr{L}_{\mathrm{MC}}$ can be seen as maximizing the symmetric version of InfoNCE. Therefore, $\mathscr{L}_{\mathrm{MC}}$ considers the RS data of two separate modalities as two views of a pair of features under the same LULC type, and trains a single-modality encoder to maximize the MI between the two modalities.

E. Multimodal Feature Matching Module

At present, the vast majority of multimodal fusion only focuses on paired samples with the same label, and does not effectively utilize multimodal samples that are not labeled the same. The unmatched samples also contain a large amount of learnable information, which will provide great assistance for RS image processing tasks with insufficient labeled samples.

To this end, we propose a multimodal matching loss function ($\mathscr{L}_{\mathrm{MM}}$) for contrastive learning. The MM loss is to train the network using multimodal samples with different labels. Firstly, define samples with the same label as positive samples, assign label 1 to these sample pairs, and there will be a total of batchsize (B) positive samples in each batch. And, for each HS image sample, select a non-paired sample that does not belong to the same label, and define these samples as negative samples. Assign a label of 0 to such sample pairs. If the same operation is performed on each LiDAR sample, there will be a total of $2B$ negative samples in each batch. $\mathscr{L}_{\mathrm{MM}}$ predicts whether a pair of multimodal samples are positive or negative samples. Using the output token of transformer as a joint representation of multimodal sample pairs, and predicting binary probabilities through a fully connected layer and a softmax to achieve a binary classification task. The expression of $\mathscr{L}_{\mathrm{MM}}$ is as follows:

$$\mathscr{L}_{\mathrm{MM}} = \mathbb{E}_{(H,L)\sim D}\,\mathrm{H}\left(y^{\mathrm{MM}}, p^{\mathrm{MM}}(H,L)\right) \tag{14}$$

where y^{MM} is the label of positive or negative samples, and $p^{\mathrm{MM}}(H, L)$ is the probability of the binary classification result.

The $\mathscr{L}_{\mathrm{MM}}$ is fundamentally a binary classification task aimed at evaluating whether the current sample aligns with another modality sample randomly selected within the batch. Although the $\mathscr{L}_{\mathrm{MM}}$ appears theoretically applicable, practical implementation reveals its simplicity as a limitation. While determining positive samples can pose challenges, defining negative samples is notably straightforward. Without specific constraints on negative sampling, numerous common instances from one modality can easily serve as negative samples for another, leading to rapid improvement in classification accuracy. However, this simplicity undermines the meaningfulness of training, so the accuracy of this task will be improved very quickly. During training, no matter how long the training lasts, it actually has no meaning.

For this, we designed a strategy to select the most challenging negative samples and then used the previous $\mathscr{L}_{\mathrm{MM}}$ to perform matching calculations on the positive and negative samples. Specifically, set the diagonal values in the similarity matrix calculated by (10) in Section III-D to 0, and set the sample values that are the same as the label to 0, that is:

$$s+ij = \begin{cases} 0, & i = j \\ 0, & \mathrm{label}(i) = \mathrm{label}(j) \\ s+ij, & \text{others.} \end{cases} \tag{15}$$

In the remaining similarity matrix, select the corresponding modal sample with the highest similarity value to each sample as the negative sample of that sample, in order to construct a negative sample pair. The samples in the negative sample pairs constructed in this way are already very, very similar. At this point, the training of $\mathscr{L}_{\mathrm{MM}}$ becomes very challenging, which can enable the model to better determine who is a true positive sample pair.

F. Classification Block

Unlike typical transformer frameworks that ultimately use an additional CLS for classification, we adopt the fusion of all tokens to obtain the final token for classification. Specifically, pre-MLP is applied to the fused features, and the features are compressed through the layernorm layer, linear layer, and softmax layer, respectively. After the softmax layer, calculate their weighted sum and input it into MLP to calculate the final classification result

$$\mathrm{pred}(i,j) = \frac{1}{\sum_{j=1}^{K}\exp(\theta_j|F)}\begin{bmatrix}\exp(\theta_1|F)\\ \cdots \\ \exp(\theta_K|F)\end{bmatrix} \tag{16}$$

where θ_j denotes the jth column of the weights in the prediction layer, while K represents the total number of LULC classes. Additionally, $\mathrm{pred}(i, j)$ corresponds to the class probabilities for the current position.

G. Network Training

Throughout the network training process, the proposed framework processes the training set samples as $\{(X_H^n, X_L^n), Y^n\}_{n=1}^{U}$. In this context, (X_H^n, X_L^n) denotes the pair of pixels associated with the same category in the HS image and LiDAR data within the training samples, while Y^n represents the corresponding labels. The total number of training samples is denoted by U. To address imbalanced quantities across different categories, we apply the weighted cross-entropy loss as the underlying loss function

$$\mathscr{L}_{\mathrm{CE}} = -\frac{1}{U}\sum_{n=1}^{U}\sum_{j=1}^{K}\omega_j \log Pr\left(Y^n = j|\left(X_H^n, X_L^n\right);\theta\right) \tag{17}$$

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where ω_j is the jth weight, $\Pr(Y^n = j|(X_H^n, X_L^n); \theta)$ is the probability that the pixel belongs to the jth class. For weights, the definition is as follows:

$$\omega_j = \frac{n_j}{U} \tag{18}$$

where n_j is the number of class j in the ground-truth training sample. To maximize the parameters, we employ an Adam optimizer with Nesterov momentum. The final loss value $\mathscr{L}$ is expressed as a combination of $\mathscr{L}_{\mathrm{CE}}$, $\mathscr{L}_{\mathrm{MC}}$, and $\mathscr{L}_{\mathrm{MM}}$

$$\mathscr{L} = \mathscr{L}_{\mathrm{CE}} + \lambda_1 \mathscr{L}_{\mathrm{MC}} + \lambda_2 \mathscr{L}_{\mathrm{MM}} \tag{19}$$

where λ_1 and λ_2 are the weight coefficients of the loss function.

IV. Experiments and Analysis

A. Dataset Description

To assess the effectiveness of the proposed framework, four HS–LiDAR datasets are utilized as experimental datasets. Below is a detailed description.

1) Houston2013 dataset: This dataset was provided by the 2013 GRSS data fusion competition. This scene was captured by airborne sensors in 2012 on the University of Houston campus and adjacent areas, with both images having a size of 349 × 1905 pixels, with a spatial resolution of 2.5 m. The HS data was captured by the CASI-1500 sensor, with 144 spectral bands and a spectral range of 0.38–1.05 μm. This dataset includes a total of 15 029 labeled samples distributed across 15 categories [44].

2) Trento Dataset: This scene was filmed in a rural area south of Trento, Italy, with both images having a size of 166 × 600 pixels, with a spatial resolution of 1 m. The HS data was captured by AISA Eagle, with 63 spectral bands and a spectral range of 0.42–0.99 μm. LiDAR data was obtained from the Optech ALTM 3100EA sensor. This dataset includes a total of 30 214 labeled samples distributed across six categories [19].

3) MUUFL dataset: This scene was captured on November 8, 2010, in the campus area of Gulf Park University of Southern Mississippi in Long Beach, Mississippi, and both images are 325 × 220 pixels. The HS data was captured by the CASI-1500 sensor, the number of spectral bands is 64, and the spectral range is 0.38–1.05 μm. The LiDAR data is obtained by the Gemini ALTM LIDAR sensor. The dataset included a total of 53 687 labeled samples, distributed across 11 categories [45], [46].

4) Augsburg Dataset: This scene was captured in the Augsburg, Germany. By downsampling the spatial resolution of the images, all images have a size of 332 × 485 pixels, with a spatial resolution of 30 m. The HS data was obtained by the HySpex sensor, with a spectral band number of 180 and a spectral range of 0.4–2.5 μm. LiDAR data was obtained from the DLR-3K system. The dataset included a total of 78 294 labeled samples, distributed across seven categories [4].

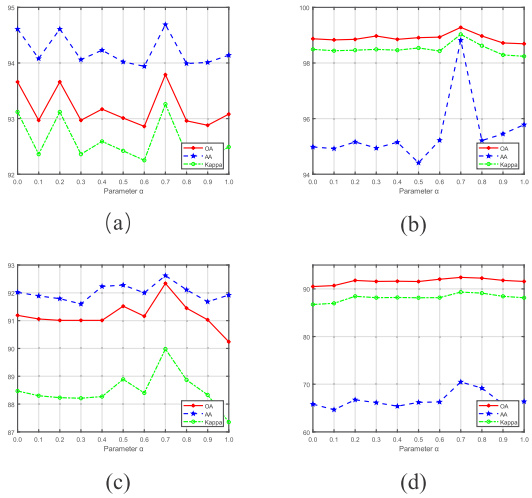


Fig. 6. Performance comparison of parameter for two similarity. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

B. Parameter Tuning

The programs are conducted using Python 3.10, with the network component developed in PyTorch. All experiments are conducted on a desktop computer featuring an Intel Core i7-13700 CPU, NVIDIA GeForce RTX 4080 GPU, and Windows 11 operating system. The HS image and LiDAR data are input to the MCFT for end-to-end learning. During the training phase, the model underwent 500 epochs with a batch size of 64, and the learning rate is 5e-4.

We performed experiments to analyze parameters that significantly influence network performance, including the balance parameter for two similarity, weights between multiple loss functions and the patch size of input data. In this study, overall accuracy (OA), average accuracy (AA), and kappa coefficient (KAPPA) are employed to assess and compare the merits and drawbacks of each method.

1) Parameter for Two Similarity Comparison: In (12), both soft similarity and hard similarity play a role in the calculation of the subsequent loss function. In order for both to play their due roles, we need to use the parameter alpha to establish a balance between them and achieve the best results. To this end, we adjusted the size of the parameters and conducted parameter experiments on four datasets. Fig. 6 shows our classification accuracy when manipulating parameters in datasets. In all the four datasets used in our experiment, we set these weight parameters to 0.7.

2) Weights Between Multiple Loss Functions Comparison: The ultimate objective function of the proposed method consists of three parts: $\mathscr{L}_{\mathrm{CE}}$, $\mathscr{L}_{\mathrm{MC}}$, and $\mathscr{L}_{\mathrm{MM}}$. Due to the fact that the main body of this method is a multiclassification task, the cross entropy loss is the most significant loss. To ensure the effectiveness of network training, we need to λ_1 and λ_2 establish a balance between them to achieve optimal results. Therefore, we change the weights of the other two loss functions and compare the results on four datasets. The OA, as depicted in Fig. 7, is evaluated by varying the weight parameters on the datasets employed in our experiment. For all four datasets, we set the weight parameters as $\lambda_1 = 0.1$ and

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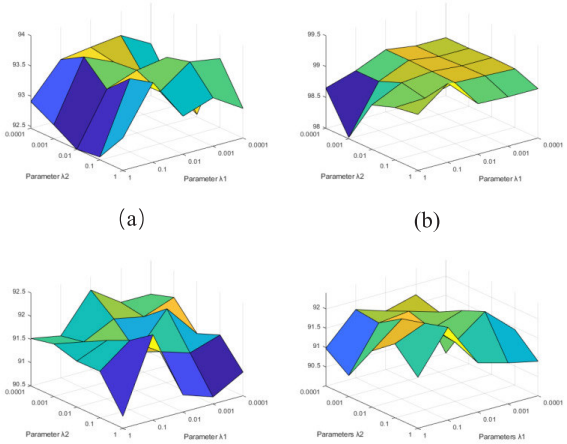


Fig. 7. Performance comparison of weights between multiple loss functions. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

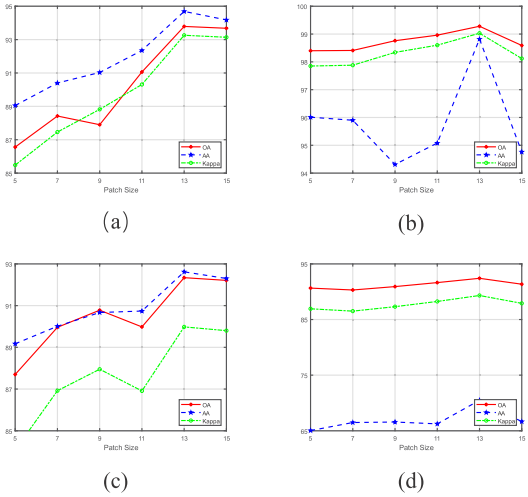


Fig. 8. Performance comparison of different input patch sizes. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

$\lambda_2 = 1$. The parameter experiment indirectly indicates that the convergence of MM loss is a very challenging task, and therefore has a greater impact on the overall loss function. Compared with $\mathscr{L}_{\mathrm{MM}}$, $\mathscr{L}_{\mathrm{MC}}$ is more important for feature extraction, but it serves the other two loss functions as a whole, so its weight is relatively small.

3) Input Patch Size Comparison: The patch size of the input to the network plays a crucial role in determining the amount of information provided to the network, and it can impact the network performance. To evaluate the influence of different patch sizes on the network, experiments were conducted and the results are shown in Fig. 8. The performance of the network was compared across patch sizes ranging from 5 to 15. It can be observed that, for all four datasets, the objective evaluation indicators for classification reach their optimal levels when the patch sizes are 13, respectively. This suggests that these particular patch sizes yield the best performance for the network in terms of classification accuracy.

4) CNN Block Number Comparison: In the feature extraction stage, the architecture of the backbone network significantly influences the overall feature representation capacity and classification accuracy of the network. While deeper CNNs can extract higher-level features through multiple convolutional layers, excessively deep architectures may result in the loss of detailed modality-specific information before the fusion process. In this section, we examine the impact of varying the number of CNN blocks in backbone feature extraction networks on the classification performance across four datasets, as summarized in Table I. The results indicate that our proposed algorithm achieved the best performance on all four datasets when utilizing three convolutional blocks. This finding is particularly relevant given that the training sample size for RS images is relatively small compared to natural images. Shallow networks often yield superior results in this context, while overly deep architectures are prone to overfitting, ultimately compromising classification accuracy.

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT NUMBERS OF CNN BLOCK

Dataset		CNN Block Number				
		1	2	3	4	5
Houston2013	OA(%)	88.73	92.59	**93.79**	93.68	92.79
	AA(%)	90.12	93.63	**94.69**	94.18	93.66
	Kappa(×100)	87.77	91.96	**93.26**	93.14	92.18
Trento	OA(%)	98.59	98.69	**99.28**	99.05	98.72
	AA(%)	93.24	95.78	**98.82**	95.97	95.45
	Kappa(×100)	98.11	98.24	**99.03**	98.72	98.29
MUUFL	OA(%)	90.72	91.52	**92.34**	91.19	91.06
	AA(%)	91.99	92.28	**92.62**	92.02	91.89
	Kappa(×100)	89.95	88.89	**89.98**	88.47	88.30
Augsburg	OA(%)	85.13	89.83	**92.41**	92.39	92.10
	AA(%)	56.97	68.46	**70.48**	66.95	61.75
	Kappa(×100)	78.67	85.65	**89.32**	89.11	88.55

C. Comparison and Analysis of Classification Performance

In the selection of comparative experiments, some SOTA methods are comprehensively considered, and algorithms based on different DL models are also considered, including TBCNN [12], CCR-Net [15], MAHiDFNet [16], CALC [20], MS2CANet [47], Glt-Net [26], ExViT [25], and MFT [28]. Among them, TBCNN, CCR-Net, MAHiDFNet, CALC, and MS2CANet are all CNNs-based methods, while Glt-Net, ExViT, and MFT are transformer-based methods. The network inputs in the same way as its publicly available code. To ensure a fair comparison of the strengths and weaknesses of different methods, publicly available code parameters were used to set the hyperparameters in the network. Furthermore, the training epochs were kept consistent across all methods.

Tables II–V show the results on four datasets under different algorithms. The best results are highlighted in bold. The results demonstrate a notable enhancement in the classification accuracy of the proposed algorithm, achieving 100% accuracy across multiple categories. Taking the Houston2013 dataset as an example, the proposed method achieved an OA of 93.79%, and the accuracy of most land cover classification reached over 85%. By utilizing sufficient feature extraction and good feature fusion, singular points in classification results can be avoided to the greatest extent possible. Meanwhile, the introduction of contrastive learning effectively utilizes sample information that does not belong to the same label, making the samples

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TABLE II
COMPARISON OF CLASSIFICATION EFFECT UNDER DIVERSE METHODS USING HOUSTON2013 DATASET

No.	Class (Train/Test)	TBCNN	CCR-Net	MAHiDFNet	CALC	MS2CANet	Glt-Net	ExViT	MFT	Propoesd
1	Health grass (198/1053)	83.10	83.00	83.10	82.24	84.62	83.10	91.55	82.81	**96.77**
2	Stressed grass (190/1064)	84.10	84.87	82.61	83.93	**95.68**	84.17	85.15	**95.68**	85.06
3	Synthetic grass (192/505)	**100.00**	**100.00**	**100.00**	93.47	**100.00**	93.07	98.61	99.41	99.41
4	Trees (188/1056)	93.03	92.14	93.37	98.86	**99.62**	93.38	98.61	98.77	96.59
5	Soil (186/1056)	**100.00**	99.81	99.72	99.72	**100.00**	99.81	**100.00**	99.91	99.53
6	Water (182/143)	99.30	95.80	**100.00**	98.60	**100.00**	**100.00**	98.60	98.60	**100.00**
7	Residential (196/1072)	92.82	**95.34**	92.26	90.21	87.03	93.66	88.90	80.04	86.75
8	Commercial (191/1053)	82.34	81.39	91.55	81.58	93.02	90.60	93.35	**96.87**	91.07
9	Road (193/1059)	84.70	84.14	86.02	48.99	89.71	**92.16**	89.52	89.24	88.48
10	Highway (191/1036)	65.44	63.22	66.31	68.15	71.43	67.18	65.54	53.19	**90.35**
11	Railway (181/1054)	88.24	90.32	95.92	95.16	98.96	83.78	95.83	92.69	**99.34**
12	Parking lot 1 (192/1041)	89.53	93.08	94.24	93.56	**98.94**	96.54	90.39	98.46	97.60
13	Parking lot 2 (184/285)	92.28	88.42	**98.60**	87.37	85.67	91.23	90.53	86.67	89.47
14	Tennis court (181/247)	96.76	96.36	**100.00**	99.60	91.90	97.98	99.60	**100.00**	**100.00**
15	Running track (187/473)	99.79	99.37	**100.00**	99.58	**100.00**	**100.00**	89.85	**100.00**	**100.00**
	OA(%)	89.98	88.15	90.06	88.91	92.52	89.50	90.62	89.97	**93.79**
	AA(%)	90.11	89.89	92.25	90.47	93.11	91.11	91.76	91.49	**94.69**
	Kappa($\times$100)	86.98	87.19	89.24	87.98	92.52	88.64	89.83	89.12	**93.26**

TABLE III
COMPARISON OF CLASSIFICATION EFFECT UNDER DIVERSE METHODS USING TRENTO DATASET

No.	Class (Train/Test)	TBCNN	CCR-Net	MAHiDFNet	CALC	MS2CANet	Glt-Net	ExViT	MFT	Propoesd
1	Apple trees (129/3905)	98.07	**100.00**	98.90	97.26	99.68	98.87	99.56	97.82	99.05
2	Building (125/2778)	95.21	98.88	97.70	**100.00**	97.49	99.14	98.13	98.60	98.13
3	Ground (105/374)	93.32	79.68	98.93	89.57	97.91	**99.47**	76.47	97.06	99.20
4	Woods (154/8969)	99.93	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	99.83
5	Vineyard (184/10317)	98.78	94.79	**100.00**	99.75	99.51	99.68	99.93	99.96	99.93
6	Roads (122/3052)	89.98	88.07	93.41	87.45	95.48	90.23	93.84	92.3	**96.76**
	OA(%)	97.92	96.57	99.03	98.16	99.05	98.67	98.80	98.72	**99.28**
	AA(%)	96.19	93.57	98.32	95.68	98.35	97.96	94.66	97.62	**98.82**
	Kappa($\times$100)	98.81	95.43	98.70	97.48	98.73	98.22	98.39	98.29	**99.03**

TABLE IV
COMPARISON OF CLASSIFICATION EFFECT UNDER DIVERSE METHODS USING MUUFL DATASET

No.	Class(Train/Test)	TBCNN	CCR-Net	MAHiDFNet	CALC	MS2CANet	Glt-Net	ExViT	MFT	Propoesd
1	Trees (100/23246)	92.07	91.10	91.71	94.12	94.72	94.07	94.03	92.66	**95.13**
2	Mostly grass (100/4270)	**89.37**	74.10	72.25	83.98	81.50	79.56	86.93	81.05	81.76
3	Mixed ground surface (100/6882)	76.11	75.08	**93.01**	75.50	81.46	78.04	75.75	75.62	85.86
4	Dirt and sand (100/1826)	93.87	91.62	94.19	89.10	**97.81**	95.29	95.24	92.06	95.45
5	Road (100/6687)	92.12	87.68	**93.57**	86.09	93.54	92.15	88.65	**93.57**	92.24
6	Water (100/466)	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
7	Building shadow (100/2233)	96.64	94.85	89.30	96.42	94.13	**96.69**	94.18	85.36	94.94
8	Building (100/6240)	94.05	94.73	96.60	95.98	95.51	95.95	96.20	**97.64**	95.32
9	Sidewalk (100/1385)	82.96	81.16	81.59	71.77	79.42	83.10	**86.35**	74.80	79.93
10	Yellow curb (100/183)	98.36	96.17	**98.91**	92.90	93.44	90.71	93.44	86.34	96.17
11	Cloth panels (100/269)	97.40	**99.63**	98.88	96.28	98.14	98.51	98.51	95.54	98.51
	OA(%)	90.18	87.74	90.99	89.55	91.66	90.77	90.63	89.52	**92.34**
	AA(%)	92.08	89.65	91.82	89.29	91.79	91.28	91.75	88.60	**92.62**
	Kappa($\times$100)	87.25	84.12	88.27	86.36	89.21	87.95	87.78	86.35	**89.98**

more distinguishable. Although not all categories achieve the best results, the significant improvement in AA also reflects a reduction in the probability of misclassification.

We evaluate the performance of various algorithms by comparing the classification result maps. Figs. 9–12 show the classification diagrams for three datasets. It can be seen that, compared with the GT map, our algorithm obtains a more accurate classification map with more pixels corresponding to GT and smoother boundaries, especially for the Trento and MUUFL datasets. This is because, compared to the Houston2013 dataset, these two datasets have simpler land classification and more continuous land distribution. In addition, our classification map does not have obvious discrete pixels. By comparing the classification maps of various algorithms, it can be found that, compared with the transformer framework, the extracted result maps have smoother and more

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TABLE V
COMPARISON OF CLASSIFICATION EFFECT UNDER DIVERSE METHODS USING AUGSBURG DATASET

No.	Class(Train/Test)	TBCNN	CCR-Net	MAHiDFNet	CALC	MS2CANet	Glt-Net	ExViT	MFT	Propoesd
1	Forest(146/13361)	95.29	93.47	93.98	94.16	**97.78**	90.82	91.83	93.47	97.77
2	Residential-Area(264/30065)	96.51	96.86	95.41	95.32	**98.38**	94.96	95.38	98.12	96.43
3	Industrial-Area(21/3830)	68.59	82.56	84.31	**86.18**	82.92	71.44	43.32	65.51	63.24
4	Low-Plants(248/36609)	82.43	84.45	92.69	95.57	91.95	92.25	91.13	94.07	**96.59**
5	Allotment(52/523)	**75.52**	44.36	63.67	0.00	62.31	69.41	41.11	27.91	66.73
6	Commercial-Area(7/1638)	15.99	0.00	7.33	6.05	8.78	7.39	**26.01**	5.25	20.82
7	Water(23/1507)	48.37	40.48	51.69	55.96	46.87	43.33	42.07	45.38	**51.76**
	OA(%)	87.31	87.82	90.76	91.46	92.17	89.13	87.82	90.86	**92.41**
	AA(%)	68.96	63.17	69.69	61.89	69.86	67.08	61.41	61.39	**70.48**
	Kappa($\times$100)	82.05	82.48	86.92	88.04	89.01	84.51	82.44	86.72	**89.32**

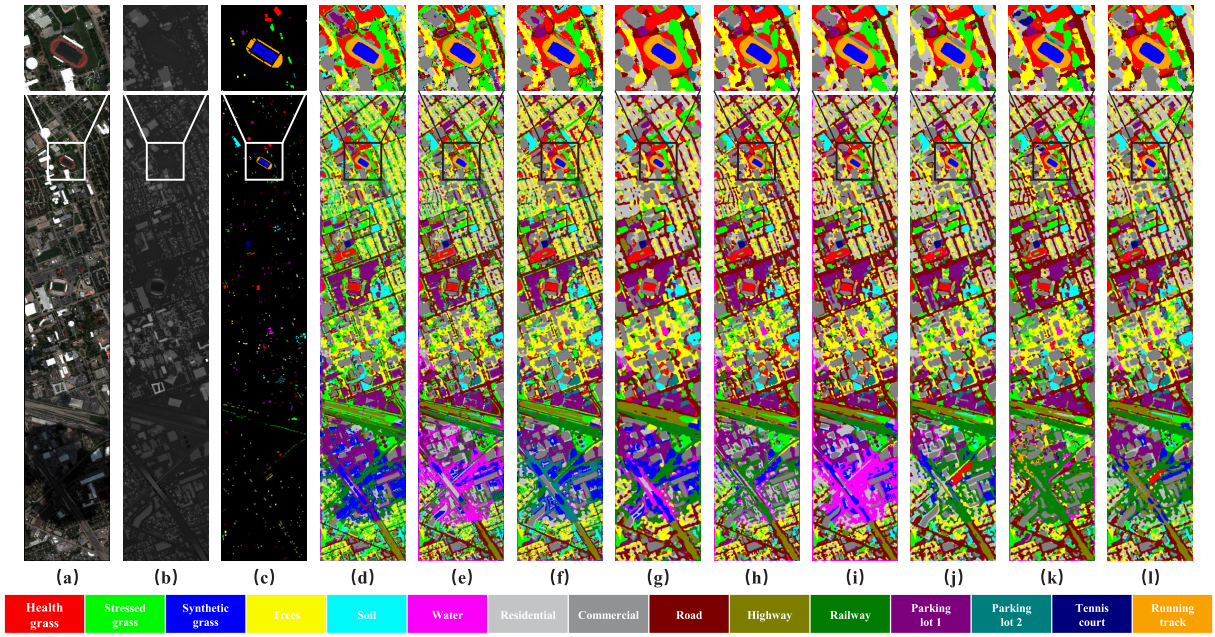


Fig. 9. Comparative analysis of classification maps for the Houston2013 dataset using different methods. (a) Pseudo-color image for HS image. (b) LiDAR-based DSM. (c) Ground-truth map. (d) TBCNN. (e) CCR-Net. (f) MAHiDFNet. (g) CALC. (h) MS2CANet. (i) Glt-Net. (j) ExViT. (k) MFT. (l) Proposed method.

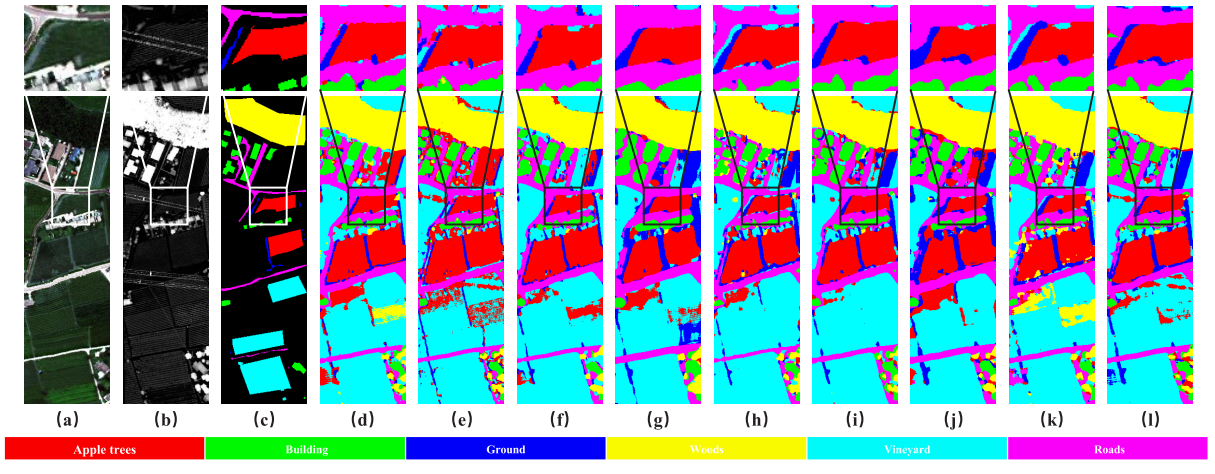


Fig. 10. Comparative analysis of classification maps for the Trento dataset using different methods. (a) Pseudo-color image for HS image. (b) LiDAR-based DSM. (c) Ground-truth map. (d) TBCNN. (e) CCR-Net. (f) MAHiDFNet. (g) CALC. (h) MS2CANet. (i) Glt-Net. (j) ExViT. (k) MFT. (l) Proposed method.

accurate classification results. This is because, compared to the traditional transformer framework, the proposed method can reduce the loss of useful information by considering stronger feature extraction and representation. Compared to the CNN

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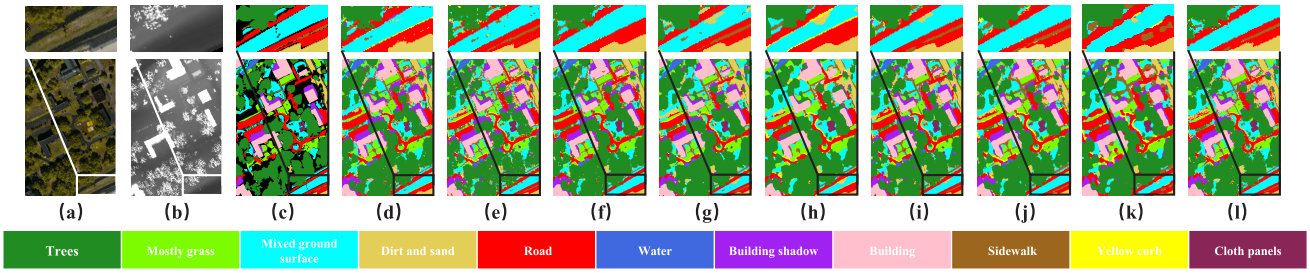


Fig. 11. Comparative analysis of classification maps for the MUUFL dataset using different methods. (a) Pseudo-color image for HS image. (b) LiDAR-based DSM. (c) Ground-truth map. (d) TBCNN. (e) CCR-Net. (f) MAHiDFNet. (g) CALC. (h) MS2CANet. (i) Glt-Net. (j) ExViT. (k) MFT. (l) Proposed method.

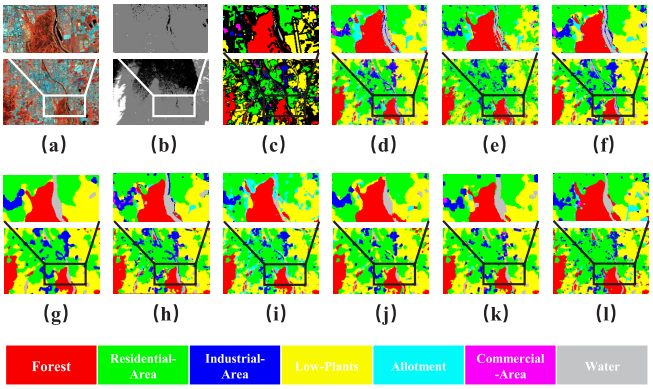


Fig. 12. Comparative analysis of classification maps for the Augsburg dataset using different methods. (a) Pseudo-color image for HS image. (b) LiDAR-based DSM. (c) Ground-truth map. (d) TBCNN. (e) CCR-Net. (f) MAHiDFNet. (g) CALC. (h) MS2CANet. (i) Glt-Net. (j) ExViT. (k) MFT. (l) Proposed method.

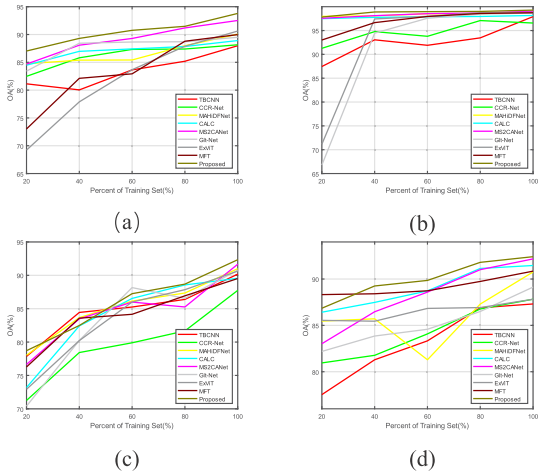


Fig. 13. Effect of classification performance across different sizes of training samples. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

methods, the proposed method fully integrates and utilizes the unique information of different modalities, which also makes the boundaries of the classification results clearer, especially in areas with more complex terrain classification.

We performed a detailed analysis by focusing on specific regions across the four datasets. In the Houston2013 dataset, we zoomed in on the stadium and its surroundings. Our proposed method, utilizing double position embedding, emphasizes the significance of the central pixel, leading to clearer boundaries and improved continuity in classifying the three types of grasses within and around the stadium. In the Trento dataset, we examined the intersection of roads and apple trees. Our method outperformed comparison algorithms, which often misclassified apple tree areas as vineyards, by employing a most challenging negative sample selection strategy in the feature matching module, thus effectively distinguishing these two similar land cover types. For the MUUFL dataset, we analyzed a street section in the bottom right corner, characterized by the overlap of four land cover categories: mixed ground surface, road, sidewalk, and dirt and sand. While some comparison methods, such as MAHiDFNet and MS2CANet, misclassified sidewalk as road, our approach clearly identifies all four categories with good continuity and minimal outliers.

In order to verify the stability of the proposed model with a small number of training samples, experiments and analyses were conducted on different training sample sizes. The detailed experimental findings are presented in Fig. 13. It is evident that the proposed method yields favorable outcomes even with a limited amount of training data. With an increasing number of training samples, the proposed framework consistently outperforms all comparison algorithms across the majority of sample sizes once classification results stabilize. In contrast to CNN-based approaches, transformer-based methods typically necessitate a larger volume of training samples to ensure result accuracy. Furthermore, it is apparent that the traditional transformer framework method is significantly impacted when the quantity of training samples is limited. However, due to the consideration of more comprehensive feature extraction, the proposed method can still maintain relatively good results in small samples, making up for the shortcomings of transformer-based methods in this regard. In the domain of multisource RS image fusion classification, acquiring sample labels is often associated with significant costs and a substantial workload. However, the proposed method exhibits the capability to achieve enhanced accuracy even with a limited number of labeled samples, highlighting its competitiveness.

In practical applications, computational efficiency and classification performance are important considerations. Table VI provides a comparison of the training and testing computational costs between the algorithm proposed in this article and two comparative algorithms. Notably, all time calculation

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TABLE VI
COMPARISON OF MODEL COMPLEXITY ON THE HOUSTON2013 DATA

Method	Parameters(M)	FLOPs(M)	Time(s)
TBCNN	6.15	0.14	2730.00
CCR-Net	0.07	0.14	291.55
MAHiDFNet	4.45	8.88	3406.85
CALC	0.90	765.62	2142.00
MS2CANet	0.69	779.04	695.66
Glt-Net	0.62	22695.66	1862.23
ExViT	0.27	3842.08	920.37
MFT	0.31	70.04	612.14
Proposed	1.66	4847.19	2086.25

experiments were executed under consistent hardware and software configurations to ensure fair comparisons. Compared with the SOTA algorithm, the proposed method has the third largest number of parameters. This is because we calculated three sets of loss functions to increase the number of learnable parameters, but it also indicates advantages in model complexity and potential overfitting mitigation. Meanwhile, the overall computation time of our method is in the middle range and still competitive, highlighting the efficiency and practical feasibility of our approach. Considering the significant improvement in classification performance results, our proposed method achieves the best balance between computational cost and performance overall.

D. Ablation Study

To evaluate the effectiveness of the proposed framework, a comprehensive set of ablation studies are conducted. These experiments systematically analyze the performance of each individual module within the framework. The results consistently demonstrate the superior performance and efficiency of our framework across all modules examined in the ablation studies. This rigorous evaluation substantiates the robustness and effectiveness of our approach in achieving the desired objectives.

1) Effect of Novel Framework: To assess the progressiveness of the proposed framework, we carry out ablation experiments on the network framework. The specific form is shown in Fig. 14. Among them, CNN serves a fine feature extraction module, transformer serves as a fine feature fusion module, linear variation serves as a simple feature extraction module, and concatenation operation serves as a simple feature fusion module. It can be seen that, for Trento datasets with relatively simple terrain distribution, feature extraction is more important, while for urban scenes where the elevation information of terrain plays a greater role, feature fusion is the more crucial step. The proposed framework considers both aspects and demonstrates superior results, underscoring the effectiveness of the framework.

2) Effect of Novel Transformer: In this section, ablation experiments are carried out to assess the performance of the enhanced transformer model. The individual contributions of the double position embedding and cross-attention modules

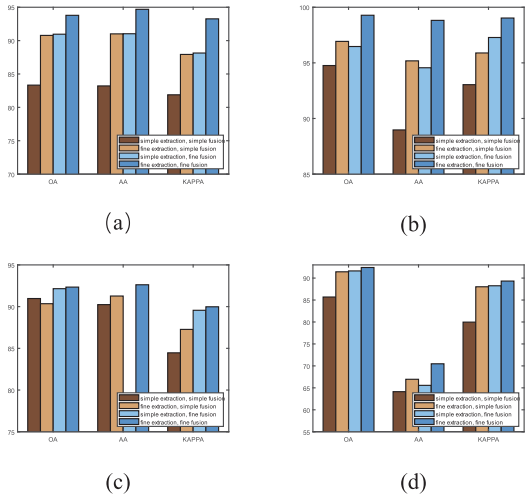


Fig. 14. Ablation study of framework. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

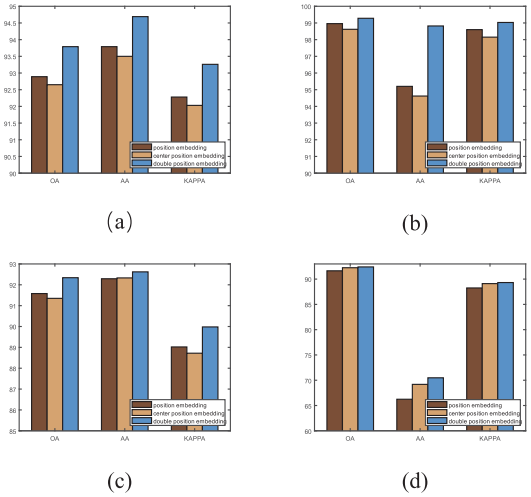


Fig. 15. Ablation study of position embedding. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

in the transformer architecture are separately evaluated and compared, with results presented in Table VII. The findings indicate a substantial enhancement in algorithm performance attributable to the application of these two modules. Furthermore, their combined utilization without mutual exclusion yielded the most optimal outcomes, thereby affirming the rationale and practicality of each module.

Additionally, a comparative analysis is conducted among the double position embedding, the center position embedding, and position embedding in the traditional ViT. The results, illustrated in Fig. 15, reveal that the position embedding and center position embedding in the traditional ViT exhibit distinct advantages and drawbacks across different datasets, owing to variations in land cover distribution within diverse scenarios. It is evident that a singular position embedding method struggles to adapt universally across all scenarios, whereas the proposed dual position embedding consistently

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TABLE VII
ABLATION STUDY OF PROPOSED MODULES

Modules		Houston2013			Trento			MUUFL			Augsburg		
double position embedding	cross attention	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
×	×	92.51	92.95	91.88	98.83	95.35	98.44	91.05	92.02	88.35	88.59	62.27	83.60
✓	×	92.65	93.50	92.03	98.62	94.62	98.15	91.58	92.29	89.02	91.43	68.21	87.77
×	✓	92.89	93.79	92.28	98.96	95.20	98.60	91.35	92.33	88.72	90.49	66.39	86.39
✓	✓	**93.79**	**94.69**	**93.26**	**99.28**	**98.82**	**99.03**	**92.34**	**92.62**	**89.98**	**92.41**	**70.48**	**89.32**

TABLE VIII
ABLATION STUDY OF CLS TOKENS

Dataset	Method	OA(%)	AA(%)	Kappa(×100)
Houston2013	With CLS	93.33	93.82	92.76
	Without CLS	**93.79**	**94.69**	**93.26**
Trento	With CLS	98.78	98.46	98.37
	Without CLS	**99.28**	**98.82**	**99.03**
MUUFL	With CLS	91.16	92.00	88.40
	Without CLS	**92.34**	**92.62**	**89.98**
Augsburg	With CLS	92.25	69.18	89.11
	Without CLS	**92.41**	**70.48**	**89.32**

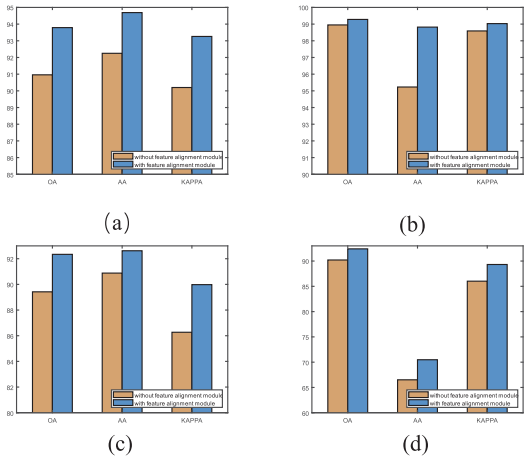


Fig. 16. Ablation study of feature alignment module. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

achieved superior results across all four datasets, demonstrating its versatility.

To address the limitation that relying solely on CLS tokens may not fully capture the comprehensive information of images, we fuse all tokens together to ensure that each token is treated equally and to preserve information to the maximum extent possible. To validate this approach, we conducted ablation experiments, with results presented in Table VIII. In traditional ViT, CLS tokens aggregate features from the entire image, allowing the model to learn global information for classification. However, in RS image processing, where training samples are often limited, depending solely on CLS tokens may lead to significant information loss due to the neglect of additional tokens. Our experimental results demonstrate the effectiveness of this strategy.

Moreover, Section III-C presented an array of feature fusion modes (depicted in Fig. 5), wherein experimental analyses were performed on diverse feature fusion techniques, and the outcomes are documented in Table IX. The results highlighted

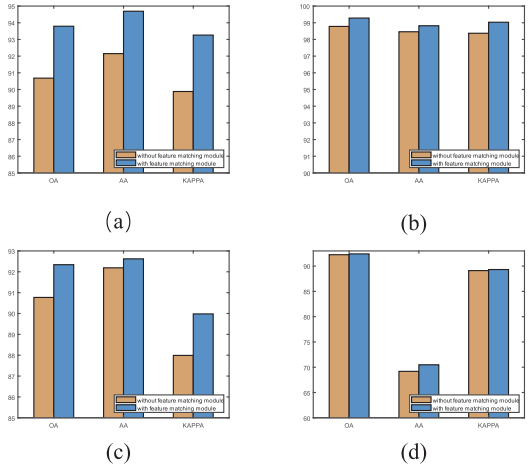


Fig. 17. Ablation study of feature matching module. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

the superior classification performance attained by the selected enhanced cross-attention module, affirming its suitability as a feature fusion mechanism for HS image and LiDAR fusion classification tasks. It was observed that traditional cross-attention predominantly prioritizes one modality, consequently compromising the information from the other mode, thus failing to comprehensively integrate multimodal data. Given the substantial disparities in the characteristics of different modalities, parameter sharing was deemed unsuitable.

3) Effect of Feature Alignment Module: The feature alignment module is designed to highlight the commonalities between identical features. To verify its effectiveness, we conduct ablation experiments on this module, and the results are shown in Fig. 16. It can be seen that the feature alignment module has a positive impact on classification accuracy, especially for more complex terrain scenes. The feature matching module is more important because the differences in terrain in the scene are very small and require more detailed differentiation.

4) Effect of Feature Matching Module: To validate the effectiveness of the feature matching module in highlighting the differences between different features, we conduct an ablation study of proposed module. The results, which are depicted in Fig. 17, clearly demonstrate that the inclusion of the feature matching module leads to improved performance, particularly for scenes exhibiting strong ground continuity. The feature alignment module plays a more critical role in avoiding the occurrence of singular points by fully considering the differences between different features.

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TABLE IX
ABLATION STUDY OF DIFFERENT FUSION MODES

Fusion Modes	Houston2013			Trento			MUUFL			Augsburg		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
(a)	92.65	93.50	92.03	98.62	94.62	98.15	91.58	92.29	89.02	91.43	68.21	87.77
(b)	88.65	90.00	87.69	98.82	95.13	98.41	87.12	88.96	83.38	91.67	69.91	88.12
(c)	90.29	91.35	89.47	98.82	95.26	98.42	90.79	91.32	88.02	91.63	67.70	88.00
(d)	**93.79**	**94.69**	**93.26**	**99.28**	**98.82**	**99.03**	**92.34**	**92.62**	**89.98**	**92.41**	**70.48**	**89.32**
(e)	90.57	91.33	89.78	98.99	94.74	98.64	90.61	90.80	87.76	89.40	62.43	84.82

V. CONCLUSION

This article introduces a novel MCFT for the fusion and classification of HS images and LiDAR data. Our approach leverages CNNs for comprehensive feature extraction from different modalities, and utilizes a transformer network as the fusion framework. We enhance the basic visual transformer framework by incorporating double position embedding and dual cross-attention, thus optimizing it for HS image and LiDAR data fusion tasks. To address the challenge of misalignment in multimodal feature interaction within transformers, we propose combining multimodal feature alignment modules to facilitate effective cross-modal learning. Additionally, we introduce a feature matching module based on challenging negative sample contrastive learning to improve the utilization of multimodal samples with varying labels during network training. Experimental results on four publicly available HS LiDAR datasets demonstrate the effectiveness of our framework. Taking the Houston2013 dataset as an example, the three objective evaluation indicators have improved by at least 1.27%, 1.58%, and 0.74%, respectively, compared to current advanced methods.

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